

Rakesh K M

E-mail: kmrakesh23@gmail.com
Phone: +91 90252 53871

LinkedIn: linkedin.com/in/kmrakesh
GitHub: github.com/Rakesh-2306

CAREER OBJECTIVE

M.Tech VLSI Design graduate (2026) with hands-on experience in analog/RF circuit design, RTL development, and ASIC implementation using industry-standard EDA tools. Proficient in Cadence Virtuoso for analog design and Synopsys toolchains for digital synthesis and physical design. Seeking opportunities in Analog/Mixed-Signal IC Design, Physical Design, or Digital IC Implementation.

EDUCATION

Vellore Institute of Technology, Chennai
M.Tech in VLSI Design 2024 – 2026

Sri Sairam Engineering College, Chennai
B.E. in Electronics and Communication 2019 – 2023

TECHNICAL SKILLS

HDLs: Verilog, SystemVerilog, TCL

Analog / RF Design: Cadence Virtuoso, RF simulation, S-parameter analysis (S_{11} , S_{12} , S_{21} , S_{22}), Noise Figure (NF) analysis, Linearity analysis (IIP3, OIP3, IMD3, P1dB), Stability analysis (K-factor, μ -factor)

Digital Design: RTL Design, FSM Design, Functional Verification

ASIC Physical Design: Synthesis, Static Timing Analysis (STA), Floorplanning, Placement, Clock Tree Synthesis, Routing

EDA Tools: Synopsys VCS, Design Compiler, ICC2, Cadence Innovus, Verilator, GTKWave, Yosys, ModelSim, Quartus

Programming: C, Python, Perl

Others: Timing Optimization, Power Analysis, DRC/LVS, Debugging

PROFESSIONAL EXPERIENCE

ITC Infotech

Associate IT Consultant Jul 2023 – Jul 2024

- Developed automation scripts for system monitoring and enterprise data processing.
- Worked with Azure, databases, and enterprise tools for system administration.

Cisco Networking Academy

Cyber Security Intern Apr 2021 – Jul 2021

- Gained experience in IP configuration and secure network setup.

ACADEMIC PROJECTS

Performance-Balanced Noise-Canceling Low Noise Amplifier Apr 2026

Designed and analyzed CG-CS, cascode, and folded-cascode LNAs in SCL 180 nm using Cadence Virtuoso; optimized folded cascode achieving 16.26 dB gain, 11 mdB NF, (S_{11} , $S_{22} < -10$ dB), and $P_{1dB} = 2.87$ dBm with unconditional stability ($K > 1$, $\mu > 1$). Implemented layout with device matching, common-centroid techniques, and parasitic-aware design; completed DRC. Designed custom square spiral inductor due to PDK limitations. Proceeding toward LVS, post-layout RC parasitic extraction, and tapeout once inductor PDK becomes available.

Ultra-Lightweight RISC-V Architecture for Anomaly Detection Nov 2025

Designed a custom RISC-V processor and implemented a full ASIC design flow. Developed RTL modules in Verilog and performed functional verification using Synopsys VCS. Executed synthesis using Design Compiler to evaluate timing, area, and power trade-offs. Performed floorplanning, placement, clock tree synthesis, and routing using ICC2 with SAED32HVT technology libraries while analyzing setup/hold timing for physical implementation.

Low Power Hybrid Full Adder for Array Multiplier Mar 2025

Designed hybrid full-adder based 4-bit and 8-bit array multipliers in 90 nm CMOS using Cadence Virtuoso. Completed full-custom layout with DRC and LVS verification. Performed post-layout RC parasitic extraction and validated delay and power, ensuring close correlation with schematic-level results.

Memristor-Based Kogge-Stone Adder Nov 2024

Designed and simulated 4-bit and 8-bit memristor-based Kogge-Stone adders to improve speed and power efficiency compared to conventional CMOS implementations. Evaluated architectural performance through delay and power analysis in Cadence Virtuoso.

PUBLICATIONS

Automated Mishap Detection and Prevention System for Vehicles, Springer ICICV 2023.
DOI: [10.1007/978-981-99-1767-9_4](https://doi.org/10.1007/978-981-99-1767-9_4)

CERTIFICATIONS

NPTEL: [System Design Through Verilog](#) — [VLSI Design Flow: RTL to GDS](#)